

Lab 1: PatientPath Descriptive Paper

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1 Introduction

1.1 Healthcare Information Exchange Challenges

The healthcare industry continues to face significant challenges in the efficient exchange of patient information, despite ongoing technological advancements. Many healthcare organizations still rely on outdated systems or fragmented electronic solutions that limit interoperability and slow down communication between providers. Notably, approximately 65% of medical record transfers still occur via fax machines, even though modern digital alternatives are widely available (Office of the National Coordinator for Health Information Technology [ONC], 2023). While some progress has been made in digital health information exchange, only a portion of hospitals routinely engage in comprehensive interoperability practices, and many still struggle to consistently share data across multiple domains (Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology, 2024). These limitations demonstrate the continued need for more efficient and reliable methods of exchanging patient information.

1.2 Limitations of Current Communication Methods

The continued reliance on fax-based communication introduces serious inefficiencies and risks into the healthcare system. Fax transmissions are inherently slow, prone to errors, and susceptible to issues such as missing pages or illegible documents (Centers for Medicare & Medicaid Services [CMS], 2020). These issues are especially problematic during patient referrals and care coordination, where incomplete or delayed information can negatively impact clinical outcomes. Studies have shown that breakdowns in referral communication often led to delayed specialist appointments, incomplete referral loops, and increased administrative burden on

healthcare staff (Patel et al., 2018). Additionally, healthcare providers frequently spend excessive time managing manual processes, including resending records and tracking down missing information, which further reduces efficiency and increases operational costs (ONC, 2020).

Despite improvements in health information technology, significant gaps remain in current systems. Many hospitals and clinics operate on heterogeneous and proprietary platforms that do not easily communicate with one another, creating barriers to seamless data exchange (Torab-Miandoab et al., 2023). Office-based physicians report high levels of difficulty when attempting to exchange data across different electronic health record (EHR) systems, with many referral partners lacking adequate data-sharing capabilities altogether (ONC, 2022). These interoperability challenges are further exacerbated in rural and under-resourced healthcare settings, where organizations are less likely to participate in routine data exchange and may face additional connectivity limitations (Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology, 2024). As a result, patient information is often incomplete, leading to redundant diagnostic testing and delays in treatment (De Swart et al., 2023).

1.3 Need For an Improved Solution

To address these issues, a modern solution must focus on improving efficiency, interoperability, security, and usability across healthcare systems. An effective system should eliminate reliance on outdated communication methods, provide seamless integration with existing EHR platforms, and support real-time data exchange between organizations.

Additionally, it must ensure the security and privacy of sensitive patient information through

robust compliance with regulations such as the Health Insurance Portability and Accountability Act (HIPAA). Scalability and accessibility are also critical, particularly for rural healthcare providers that may experience connectivity challenges. Without addressing these core requirements, existing solutions will continue to fall short in meeting the needs of modern healthcare environments.

PatientPath is introduced as a comprehensive solution designed to overcome these challenges. PatientPath is a secure, centralized, web-based platform that enables efficient medical record sharing and referral coordination across healthcare organizations. Rather than replacing existing EHR systems, PatientPath functions as an intermediary that bridges gaps between disparate systems, allowing providers to exchange information seamlessly. By combining secure digital record exchange, interoperability standards such as FHIR and HL7, automated referral workflows, and offline synchronization capabilities, PatientPath provides a scalable and practical approach to modernizing healthcare communication. Through these features, PatientPath aims to reduce delays, improve data accuracy, and enhance overall patient care outcomes. This paper will demonstrate the key characteristics of PatientPath through a structured prototype-based approach, highlighting how the system addresses real-world healthcare communication challenges.

2 PatientPath Product Description

PatientPath is a secure, centralized, web-based healthcare platform designed to improve the transfer of medical records and coordination of patient referrals between healthcare organizations. The system was created to address the ongoing inefficiencies associated with fax-based communication and fragmented electronic health record (EHR) systems. Many healthcare

providers continue to experience delays, incomplete patient information, and administrative burdens when attempting to exchange records across organizations. PatientPath aims to reduce these issues by providing a unified platform that supports secure digital communication and streamlined referral workflows.

Unlike traditional EHR systems, PatientPath is not intended to replace existing healthcare software platforms. Instead, it functions as a bridge between organizations and their current systems, allowing providers to exchange patient information more efficiently regardless of which EHR platform they use. The platform focuses on improving interoperability, increasing workflow efficiency, and strengthening communication between hospitals, clinics, specialists, and rural healthcare providers. Through this approach, PatientPath seeks to improve care coordination, reduce delays in treatment, and enhance the overall patient experience.

The primary objective of PatientPath is to modernize healthcare communication through secure digital record exchange and automated referral management. The system emphasizes accessibility, security, and scalability while maintaining compatibility with industry standards such as FHIR and HL7. By simplifying the process of sharing patient information across organizations, PatientPath provides healthcare providers with a more reliable and efficient alternative to outdated transfer methods.

2.1 Key Product Features and Capabilities

Secure Real-Time Digital Record Exchange

PatientPath includes a secure real-time digital record exchange system that replaces traditional fax-based communication with instant electronic document transfer. Healthcare providers are able to upload, transfer, and retrieve patient records digitally, reducing delays caused by manual fax transmission. This feature minimizes common issues such as missing pages, illegible documents, and multi-day delivery delays while improving the overall efficiency of healthcare communication. The prototype should allow an authorized provider to upload a patient record, associate it with the appropriate patient, and make the record available to an authorized receiving provider without requiring a fax-based transfer. A successful transfer should preserve the complete uploaded record and make its availability visible through the system.

FHIR / HL7 Interoperability

Interoperability is one of the primary design goals of PatientPath. The platform supports FHIR R4 and HL7 healthcare data exchange standards, allowing it to integrate with multiple electronic health record systems. Because healthcare organizations frequently use different software platforms, interoperability is critical for enabling consistent communication between providers. PatientPath's EHR-agnostic architecture allows organizations to adopt the platform without replacing their existing systems.

Automated Referral Workflow and Status Tracking

PatientPath includes an automated referral workflow system that improves coordination between referring providers and specialists. Providers can create referrals, share patient records, and monitor referral status updates through a single interface. Automated validation checks ensure that required patient information is complete before transmission, reducing the need for

repeated phone calls and manual follow-up procedures. Referral tracking also improves transparency throughout the referral process. The referral workflow should require all necessary referral information before a referral can be submitted and should maintain an identifiable status for each referral throughout the process. Providers should be able to determine whether a referral has been created, received, or updated through the PatientPath interface.

Security and HIPAA Compliance

Because healthcare organizations manage highly sensitive patient information, security is a major priority within PatientPath. The platform uses end-to-end encryption to protect patient data during both transmission and storage. Additional security measures include multi-factor authentication (MFA), role-based access control (RBAC), and audit logging systems that track all record access and transfer activity. These protections help healthcare organizations maintain HIPAA compliance while reducing privacy and security risks. Access to protected patient information should be limited according to the authenticated user's role, and record-access and transfer activities should produce an audit-log entry.

Offline Sync Resilience for Rural Clinics

PatientPath also addresses connectivity challenges experienced by rural healthcare providers. Many smaller clinics experience unstable internet connectivity that can interrupt cloud-based healthcare systems. To improve reliability, PatientPath includes offline caching and synchronization capabilities that allow providers to continue working during network outages. Once connectivity is restored, the system automatically synchronizes stored records with the platform.

2.2 Major Product Components

Presentation Layer

The presentation layer of PatientPath provides the user-facing interfaces used by healthcare providers, patients, and administrators. The Provider Dashboard allows healthcare professionals to manage referrals, access patient records, and monitor transfer status updates. The Patient Portal enables patients to review shared records and manage consent preferences. Additional interfaces include login systems, notification dashboards, and administrative management tools designed to improve usability for multiple user roles.

Application Layer

The application layer contains the core services responsible for system functionality and communication between components. Major services include the Secure Record Exchange Service, Identity and Access Management (IAM), Patient Record Management, and the FHIR/HL7 Integration Service. An API Gateway and web server coordinate communication between users, healthcare systems, and stored patient data. Together, these services support secure interoperability and workflow automation throughout the platform.

Data Layer

The data layer is responsible for storing and managing healthcare information securely. PatientPath uses encrypted document storage for medical records and a PostgreSQL database for managing users, organizations, and referral information. Additional stored information includes

audit logs, consent records, and normalized FHIR data. This architecture supports secure access management while maintaining system scalability and reliability.

Infrastructure and Scalability

PatientPath is designed as a cloud-hosted platform compatible with major providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). The system is intended to support horizontal scaling and high availability in order to maintain reliable operation as healthcare networks expand. Redundancy and failover protections are included to improve uptime and reduce service interruptions.

The major components of PatientPath work together to support the medical record and referral workflow. A healthcare provider begins the process through the presentation layer by accessing the Provider Dashboard and creating a referral or selecting patient records for transfer. The request is then passed to the application layer through the API Gateway, where services such as Identity and Access Management, Patient Record Management, and the Secure Record Exchange Service process the request and verify that the user has appropriate access. When interoperability is required, the FHIR/HL7 Integration Service provides a standardized method for exchanging healthcare information between systems. The data layer stores and retrieves the necessary patient, referral, consent, and audit information using the PostgreSQL database and encrypted document storage. These components operate together within the system infrastructure to provide a continuous workflow from user interaction to secure data processing, storage, and exchange.

3 Identification of Case Study

The PatientPath platform is being developed to support healthcare providers that experience difficulties transferring patient information between organizations. The primary case study focuses on Dr. Sarah Chen, a primary care physician working at a mid-sized rural clinic in rural Virginia with limited information technology infrastructure. Dr. Chen's clinic uses an older EHR system that cannot directly exchange patient records with external healthcare organizations. One of her patients, Robert Mills, is a 67-year-old male recently diagnosed with a cardiac condition requiring referral to a cardiologist at a regional hospital located approximately 45 miles away.

The current referral process used by Dr. Chen's clinic relies heavily on fax-based communication. Staff members must manually send more than fifteen pages of patient records to the specialist's office, often resulting in delays caused by missing pages or incomplete information. The specialist's office frequently contacts the clinic requesting additional documentation, which extends the referral timeline and delays patient treatment. As a result, Robert's specialist appointment may be delayed by an entire week due to communication failures and inefficient record transfers.

PatientPath is intended to address these challenges by providing secure digital record exchange, automated referral management, and interoperability between different EHR systems. Using PatientPath, Dr. Chen can create referrals, select records for sharing, and grant specialist access through a single workflow. The specialist's office receives instant notifications and immediate access to complete patient documentation without relying on fax machines or manual follow-up calls. Through this case study, PatientPath demonstrates how improved

interoperability and workflow automation can reduce delays and improve healthcare communication.

4 Prototype Description

The PatientPath prototype is designed to demonstrate the core functionality and workflow concepts of the full PatientPath platform. While the final product is intended to operate as a scalable healthcare communication system, the prototype focuses primarily on illustrating the major features and user interactions that address interoperability and referral management challenges. The prototype serves as a proof of concept that demonstrates how secure digital referrals and medical record sharing can improve communication between healthcare providers.

The prototype will include several major features from the final system, including secure digital record exchange, referral creation, referral status tracking, and role-based user interfaces for providers and administrators. Users will be able to simulate the process of creating referrals, uploading patient records, granting organizational access, and viewing referral progress through the platform dashboard. Simplified authentication and notification systems will also be included to demonstrate overall workflow functionality.

Although the prototype demonstrates many of the system's primary workflows, it differs from the final product in several important ways. The prototype will not include live integration with commercial EHR platforms such as Epic or Oracle Healthcare. Instead, interoperability features will be simulated to demonstrate how information exchange would function in a production environment. Similarly, the prototype will not include full-scale cloud infrastructure, advanced failover systems, or complete HIPAA compliance certification processes.

The purpose of the prototype is to validate the usability, workflow efficiency, and overall feasibility of the PatientPath platform before full-scale development. By focusing on core healthcare communication challenges and demonstrating practical workflow improvements, the prototype provides a strong foundation for future expansion into a production-ready healthcare interoperability system.

5 Glossary

EHR — Electronic Health Record

HIPAA — Health Insurance Portability and Accountability Act

FHIR — Fast Healthcare Interoperability Resources (standard for health data exchange)

HL7 — Health Level Seven (international healthcare data standards organization)

RBAC — Role-Based Access Control

MFA — Multi-Factor Authentication

API — Application Programming Interface

REST — Representational State Transfer (architectural style for APIs)

HIE — Health Information Exchange

PCP — Primary Care Provider

FQHC — Federally Qualified Health Center

CI/CD — Continuous Integration / Continuous Deployment

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